

Alteration of Oligodendrocyte Functional Mechanics By Neurodegenerative Disease

Kamilah Y. Amen, Nicholas J. Braun, Patrick W. Alford

Department of Biomedical Engineering, University of Minnesota

TRIO
RONALD E. MCNAIR POST-BACCALAUREATE ACHIEVEMENT PROGRAM

BACKGROUND

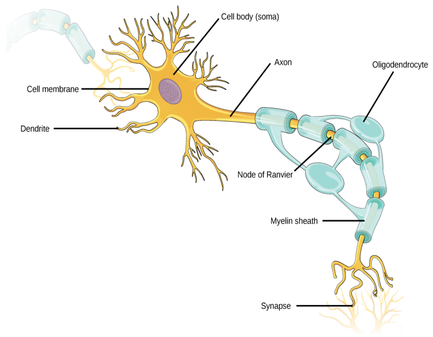


Figure 1. A diagram of a neuron, oligodendrocyte, and dendritic spine
A.This image was taken from *Neurons and Glial Cells: Figure 2* by OpenStax[2]

- Neurodegenerative diseases affect one billion people worldwide[1] and alter the function and properties of the cells they affect
- Both amyloid beta-1-40 (Aβ-1-40) & amyloid beta-1-42 (Aβ-1-42), and alpha-synuclein (α-synuclein) are proteins overactive in Alzheimer's and Parkinson's disease, respectively
- Neurons in the central nervous system are targets of these proteins
- Neurons are protected and supported by surrounding glial cells
- Oligodendrocytes are glial cells that produce myelin sheaths, wrapping around neuronal axons, protecting the axons and strengthening action potentials
- The interactions between the aggregated proteins and oligodendrocytes and the effects on the traction force are unknown
- **We use traction force microscopy (TFM) and immunofluorescent staining (IF) to explore how disease related extracellular factors alter cellular force generation and morphology**

EXPERIMENTAL METHODS

Cell Substrate Construction

- Ultra violet ozone (UVO) radiation was applied to the 25mm coverslips for 8 minutes to sterilize the surface
- Bind-silane applied onto surface of 25mm coverslips until dry
- 15mm coverslips were cleaned, plasma treated, and treated with poly-D lysine (PDL) for 30 minutes
- 2.5 kPa polyacrylamide gel was fused with fluorescent beads that measured the movement of the cells' appendages in TFM
- 10 µL of gel was placed onto 25mm coverslips and covered with PDL treated 15mm coverslips to cure for 1 hour
- After 15mm coverslips are removed, gels remain attached to 25 mm coverslips and treated with 4% bovine serum albumin (BSA)

Figure 2. Fluorescent polyacrylamide gel application schematic

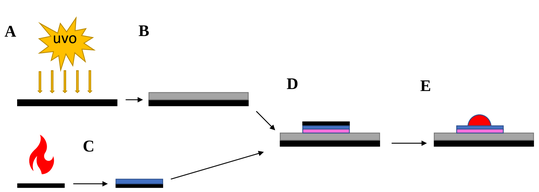


Figure 2 steps:
A). 25mm coverslip treated with UVO radiation
B). Bind-silane applied until dry on coverslip
C). 15mm coverslip plasma treated and soaked in PDL for 30 minutes
D). Gel applied and cured
E). 15mm coverslip removed and cells seeded onto gel

EXPERIMENTAL METHODS

Cell Culture

- Oligodendrocyte cells were harvested from neonatal rat pups by Braun, N
- Cells were kept in an incubator at 37˚C and in standard culture medium; media was changed every two days
- Cells were treated with 1µM of proteins (Aβ-1-40, Aβ-1-42, and α-synuclein) 24 hours before TFM was performed
- Cells were seeded onto the gels and allowed to adhere for 24 hours

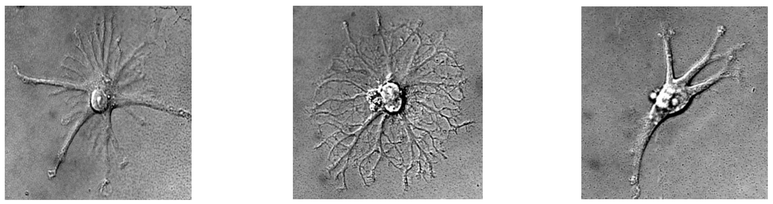


Figure 3. Oligodendrocytes in culture

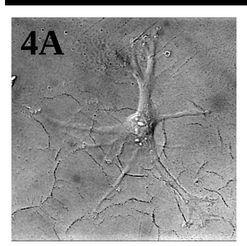
Traction Force Microscopy (TFM)

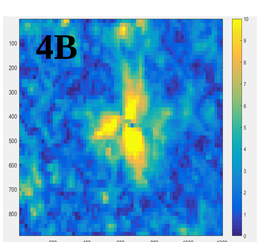
- All experiments were performed using an Olympus IX81ZDC confocal[3,4] and Zeiss Axio Observer[5] inverted microscope at 40X magnification
- Gel-embedded fluorospheres were imaged with the cell intact
- Cell was lysed from the gel using sodium dodecyl sulfate (SDS) and gel embedded beads were imaged again
- Cell traction forces were measured using previously determined methods[3,4]

Immunofluorescent (IF) Staining

- Cells were fixed with 4% paraformaldehyde for 5 minutes
- Phalloidin and DAPI, in a 1:200 dilution with 0.5% BSA in 1X phosphate-buffered saline (PBS), were used to label actin and chromatin, respectively
- Cells were imagined using an Olympus IX81ZDC confocal[3,4] and Zeiss Axio Observer[5] inverted microscope at 40X magnification

RESULTS





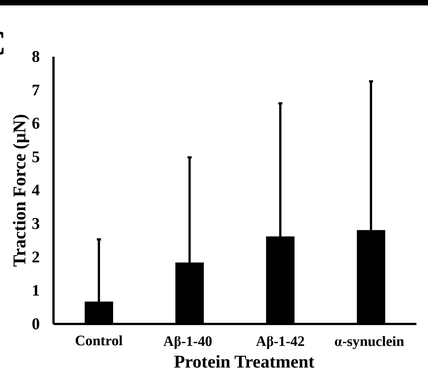


Figure 4. Results for TFM
A). Oligodendrocyte under 1µM treatment of aβ-1-40
B). Traction stress heat map of oligodendrocyte in figure 4A
C). Protein treatment vs traction force average; (n=38, 26, 3, 23; error bars = standard deviations)

RESULTS

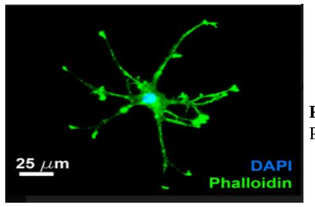
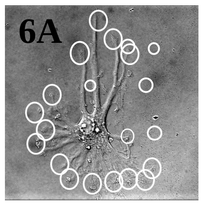
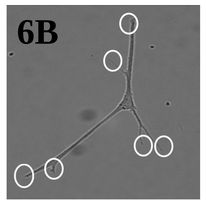


Figure 5. Oligodendrocyte stained with DAPI (blue) and Phalloidin (green). Courtesy of Nicholas J. Braun





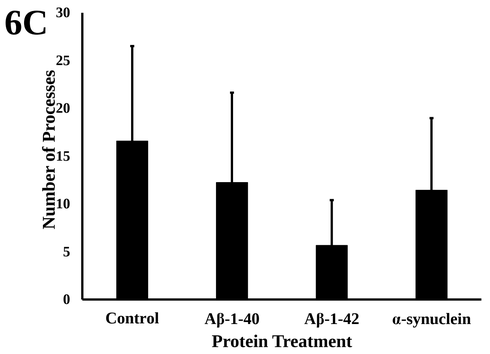


Figure 6. Results for number of processes
A). Oligodendrocyte with no protein treatment
B). Oligodendrocyte with 1µM α-synuclein treatment
C). Protein treatment vs number of processes on oligodendrocytes; (n=38, 26, 3, 23; error bars = standard deviations)

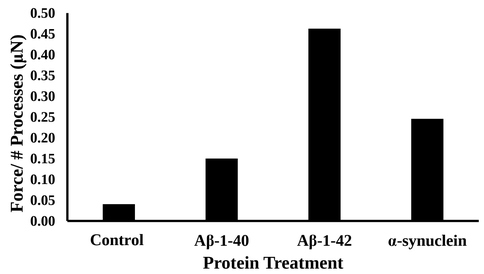


Figure 7. Protein treatment and the corresponding force/process value; (n=38, 26, 3, 23; error bars = standard deviation)

DISCUSSION

- Glial cells provide the substrate on which neurons exist and function
- Our data suggest that Aβ-1-40, Aβ-1-42, and α-synuclein alter the traction force generated by oligodendrocytes
- This increase in force could indicate disturbances in neuronal function and mediate disease progression

REFERENCES

1. Bertolote, Jose Manoel. (2007, Feb. 27). Neurological disorders affect millions globally: WHO report. World Health Organization. WHO Geneva. Retrieved from http://www.who.int/mediacentre/news/releases/2007/pr04/en/
2. OpenSrax CNX. Neurons and Glial Cells-Bioloigy-OPenStax CNX. *OpenStaX, Biology* (2015). at https://cnx.org/contents/c9j4p0aj@3/Neurons-and-Glial-Cells
3. Steucke, K.E. et al. (2017, April 17). Empircally Determined Vascular Smooth Muscle Cell Mechano-Adaptation Law. *J. Biomech. Eng.* **3**. Article Number: BIO-16-1491.
4. Win, Z. et al. (2017, April 13). Cellular Microbiaxial Stretching to Measure a Single-Cell Strain Energy Density Function. *J. Biomech. Eng.* **3.** Article Number: BIO-16-1446.
5. Wood, D.K. et al. (2010, Mar. 31). Single cell trapping and DNA damage analysis using microwell arrays. Proc Natl Acad Sci USA 107, 10008.